MAT2540, Quiz3, Spring2026

ID:_____ Name:_____

- This quiz consists of 1 question for a total of 10 points.
- You have 15 minutes to complete the quiz.
- Show all work and justify your answers.
- Wishing you success.

1. Given a nth-term sequence defined by

$$a_0 = 0, a_1 = 0, a_2 = 1, \text{and } a_n = a_{n-1} + a_{n-2} + a_{n-3}, \text{ for } n = 3, 4, 5, \ldots.$$

a) Devise a recursive algorithm to find this sequence:

procedure *Tribonacci*(n: non-negative integer)

if n=0 then return 0
else if n=1 then return 0
else if n=2 then return 1
else return Tribonacci (n-1) + Tribonacci (n-2) + Tribonacci (n-3)

{ output is Tribonacci (n) }

b) Devise an iterative algorithm to find this sequence:

procedure *iterative Tribonacci*(n: nonnegative integer)

if n=0 then return 0
else if n=1 then return 0
else if n=2 then return 1
else
  x:=0, y:=0, z:=1
  for i:=1 to n-1
    w=x+y+z ; x:=y ; y:=z ; z:=w
  return z

{ output is n^{th} of the Tribonacci number }